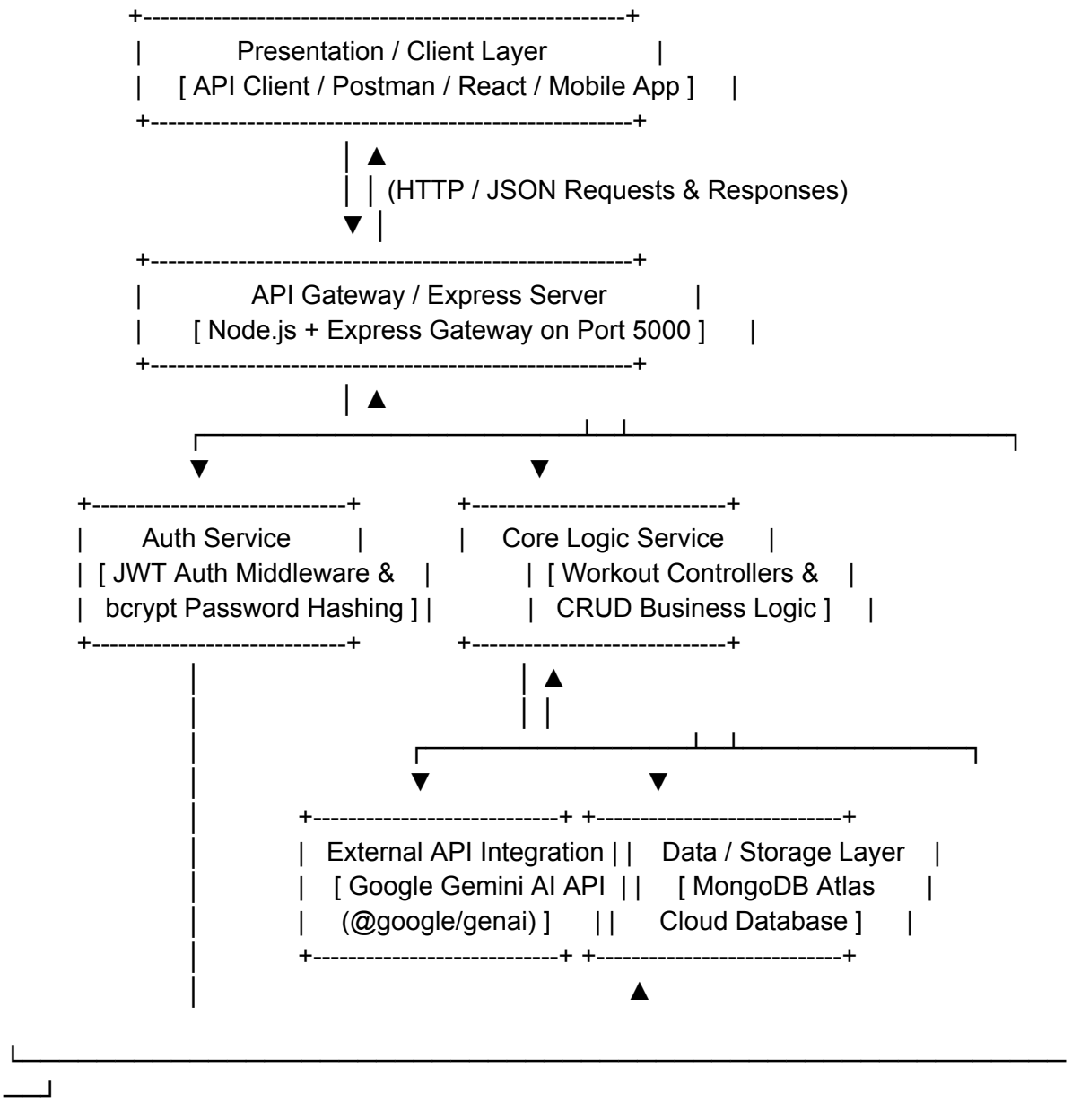
Solution Architecture

SMARTBRIDGE - Solution Architecture

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 5 Marks

Solution Architecture Diagram Structural Layout



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Serves as the entry point for users to log in, input workout metrics, search history, and receive AI workout recommendations.	Postman, Thunder Client, React.js / Mobile Client
API Gateway	Central HTTP server listener that routes requests, handles CORS policy, parses JSON, and forwards calls to controllers.	Express.js Gateway (server.js, app.js)
Auth Service	Verifies user credentials, hashes passwords, generates signed JWTs, and authorizes protected routes via middleware.	jsonwebtoken, bcryptjs, authMiddleware.js
Core Logic Service	Executes workout CRUD logic, performs search queries, calculates stats, and formats JSON API output.	Express Controllers (workoutController.js, aiController.js)
External API Integration	Communicates with Google Gemini Generative AI to generate tailored weekly workout plans and progress insights.	Google Gemini API (@google/genai, gemini-2.5-flash)
Database Layer	Stores persistent user credentials, profile records, and complete workout logs in structured NoSQL collections.	MongoDB Atlas Cloud, Mongoose ODM (User.js, Workout.js)